

# Airline Delay Analysis – Project Log

## Project Information

**Project Title:** Airline Delay Analysis (2003–2022)

**Objective:**

Analyze airline delay patterns across U.S. airlines and airports, identify the primary causes of delays, evaluate airline and airport performance, and explore seasonal and long-term trends using SQL and Tableau.

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## Phase 1: Dataset Selection

### Date

June 4, 2026

### Activity

Selected the Airline Delay Cause dataset from Kaggle.

### Reason for Selection

The dataset provides detailed information on flight operations, delays, cancellations, diversions, airlines, airports, and delay causes. It was selected because it supports SQL practice, business analysis, dashboard development, and data storytelling.

### Initial Learning

The dataset contains:

- Year and month information
- Airline and airport identifiers
- Flight volume metrics
- Delay counts and delay minutes
- Delay causes (Carrier, Weather, NAS, Security, and Late Aircraft)

Initially, I was unfamiliar with several delay-related fields, particularly the delay count (CT) columns. After reviewing the dataset documentation, I confirmed that these fields represent delay counts categorized by delay cause.

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## Phase 2: Exploratory Data Analysis (EDA)

### Objective

Understand the structure, scale, and quality of the dataset before beginning detailed analysis.

### SQL Analysis Performed

- Counted total records
- Identified date range
- Counted distinct airlines
- Counted distinct airports

### Findings

- Total Records: 318,017
- Time Period: 2003–2022
- Airlines: 33
- Airports: 420

### Insight

The dataset provides broad coverage of the U.S. airline network across a 20-year period, making it suitable for trend analysis, airline performance evaluation, airport comparisons, and delay-cause investigation.

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## Phase 3: Data Quality Assessment

### Missing Value Investigation

Reviewed records containing missing carrier information.

### Findings

- Four records contained missing carrier data.
- All four records occurred during August 2008.
- The issue appeared isolated and did not indicate a broader data quality problem.

## Conclusion

The missing values were determined to be minor and unlikely to affect analysis results.

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## Duplicate Check

### Analysis Performed

Checked for duplicate airline-airport-month observations.

### Findings

No duplicate records were identified.

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## Data Validation

Verified:

- Month values ranged from 1–12
  - Year values ranged from 2003–2022
  - Key operational fields contained valid values
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# Phase 4: Data Cleaning

## Cleaning Activities

- Checked for missing values across operational columns.
- Investigated incomplete records.
- Removed records containing missing critical flight and delay metrics.
- Preserved the original raw dataset.
- Created a cleaned analysis table.

## Output

Created cleaned analysis table:

### **Airline\_Delay\_Clean**

This table was used for all subsequent analysis and visualization work.

### **Key Learning**

Data quality validation is essential before analysis because even a small number of incomplete records can affect summary metrics and business conclusions.

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## **Phase 5: Airline Performance Analysis**

### **Objective**

Evaluate airline performance using both total delay volume and average delay burden.

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### **Total Delay Minutes by Airline**

#### **Analysis Performed**

Calculated cumulative delay minutes for each airline.

#### **Findings**

Airlines with the highest total delay minutes included:

- Southwest Airlines
- American Airlines
- SkyWest Airlines
- Delta Air Lines
- United Air Lines

#### **Observation**

Total delay minutes alone can be misleading because larger airlines naturally operate more flights.

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# Average Delay Minutes per Flight

## Analysis Performed

Calculated:

Average Delay Per Flight = Total Delay Minutes ÷ Total Flights

Applied a minimum flight threshold to avoid distortion from very small airlines.

## Findings

Highest average delay per flight:

- ExpressJet Airlines LLC – 16.67 minutes
- Allegiant Air – 15.89 minutes
- JetBlue Airways – 15.61 minutes

## Key Learning

Average delay per flight provides a more meaningful measure of operational efficiency than total delay volume.

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# Phase 6: Delay Cause Analysis

## Objective

Determine which delay categories contribute most to overall airline delays.

## Analysis Performed

Aggregated total delay minutes for:

- Carrier Delay
- Weather Delay
- NAS Delay
- Security Delay
- Late Aircraft Delay

## Findings

Delay causes ranked from highest to lowest:

1. Late Aircraft Delay
2. Carrier Delay
3. NAS Delay
4. Weather Delay
5. Security Delay

### **Insight**

Late aircraft delays consistently represented the largest source of delay minutes across the industry, while security-related delays contributed only a negligible portion of total delays.

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## **Phase 7: Airport Performance Analysis**

### **Objective**

Identify airports with the highest delay burden.

### **Analysis Performed**

Calculated average delay per flight by airport.

### **Findings**

Newark Liberty International Airport recorded the highest average delay per flight.

Other major New York-area airports also ranked among the poorest-performing airports, suggesting persistent congestion challenges.

### **Insight**

Airport congestion appears to play a significant role in delay performance, particularly at major hub airports.

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## **Phase 8: Trend Analysis**

# Yearly Delay Trends

## Objective

Evaluate how delay performance changed over time.

## Analysis Performed

Calculated annual totals for:

- Flights
- Delay minutes
- Average delay per flight

## Findings

- Delay performance worsened during portions of the mid-2000s.
- Delay levels fluctuated throughout the following decade.
- Delay minutes dropped sharply during 2020.
- Delays increased again during 2021–2022.

## Insight

The sharp decline during 2020 should be interpreted cautiously because it coincided with the COVID-19 pandemic and a substantial reduction in flight activity.

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# Monthly Delay Trends

## Objective

Identify seasonal patterns in airline delays.

## Analysis Performed

Calculated monthly flight volume, delay minutes, and average delay per flight.

## Findings

Highest delay months:

- June
- July

- August
- December

Lowest delay months:

- September
- October
- November

## Insight

Airline delays demonstrate clear seasonality, with summer travel periods and holiday travel creating increased operational pressure.

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# Phase 9: Business Question Development

After completing exploratory analysis and data cleaning, I refined the project scope and developed five business-focused questions designed to uncover operational insights from the dataset.

## Final Business Questions

1. What delay causes contribute most to industry-wide delays?
2. Which airlines have the highest delay burden relative to flight volume?
3. Which airports act as operational bottlenecks?
4. How do delay patterns vary across months and seasons?
5. How have airline delays evolved from 2003–2022?

## Key Learning

Creating focused business questions helped transform exploratory findings into actionable analyses and provided a clear framework for dashboard development.

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# Phase 10: SQL Analysis & Summary Table Creation

To answer the final business questions, dedicated SQL queries were created for each analysis area.

## Tables Created

- Q1\_Delay\_Causes
- Q2\_Airline\_Delay\_Rates
- Q3\_Airport\_Bottlenecks
- Q4\_Monthly\_Delay\_Patterns
- Q5\_Yearly\_Delay\_Trends

## Process

- Built SQL queries to answer each business question.
- Validated results against earlier exploratory findings.
- Created analysis-ready summary tables.
- Exported all tables from SQLite as CSV files.

## Key Learning

Creating dedicated summary tables simplified visualization development and improved dashboard performance.

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# Phase 11: Tableau Dashboard Development

## Objective

Transform SQL outputs into visualizations that communicate findings effectively to non-technical stakeholders.

## Data Preparation

Imported all exported CSV files into Tableau Public.

## Visualizations Created

### Q1 – Major Causes of Airline Delays

- Pareto-style bar chart showing the contribution of each delay cause.

## **Q2 – Top Airlines by Delay Rate**

- Ranked comparison of airlines based on delayed flights as a percentage of total flights.

## **Q3 – Airport Traffic Volume vs Average Delay**

- Scatter plot comparing airport traffic volume and average delay minutes to identify operational bottlenecks and explore the relationship between traffic volume and delay severity.

## **Q4 – Seasonal Delay Patterns**

- Time-based visualization showing how delay performance varies throughout the year.

## **Q5 – Airline Delay Trends (2003–2022)**

- Line chart illustrating long-term changes in delay performance across the industry.

## **Tableau Skills Practiced**

- Data source management
- Worksheet design
- Filtering and sorting
- Label formatting
- Axis customization
- Color encoding
- Dashboard storytelling

## **Key Learning**

A visualization should not only display data but also clearly communicate the business question being answered and the insight being delivered.

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# **Skills Demonstrated**

## **SQL**

- SELECT
- WHERE
- GROUP BY
- HAVING
- ORDER BY

- Aggregate Functions
- Data Validation
- Data Cleaning

## **Analytics**

- Exploratory Data Analysis (EDA)
- Trend Analysis
- Root Cause Analysis
- KPI Development
- Business Question Formulation

## **Visualization**

- Tableau Dashboard Development
  - Comparative Analysis
  - Trend Visualization
  - Data Storytelling
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# **Project Outcome**

This project successfully identified the primary causes of airline delays, highlighted differences in airline and airport performance, revealed seasonal patterns in delay behavior, and examined how airline delays evolved over a twenty-year period.

The project followed a complete analytics workflow from dataset selection and data cleaning through SQL analysis, business question development, dashboard creation, and business insight generation.